

EDA Commands Cheatsheet

Python Pandas | SQL | Visualization

Data Inspection

```
df.head(), df.tail()          # First/last N rows
df.info()                     # Data types, non-null counts
df.shape                      # Rows, columns
df.dtypes                     # Column data types
df.describe(include='all')    # Numeric + categorical summary
```

Missing Data

```
df.isnull().sum()             # Count nulls per column
df.isnull().sum() / len(df) * 100 # Null percentage
df.dropna(), df.fillna(value)  # Handle missing values
```

Duplicates & Unique Values

```
df.duplicated().sum()         # Count duplicates
df.drop_duplicates()          # Remove duplicates
df['col'].unique()             # Unique values
df['col'].nunique()            # Count unique
df['col'].value_counts()       # Frequency distribution
```

Statistical Summary

```
df.mean(), df.median(), df.std() # Central tendency
df.min(), df.max()               # Range
df.quantile([0.25, 0.5, 0.75])  # Percentiles
df.corr()                        # Correlation matrix
df['col'].skew(), df['col'].kurtosis() # Distribution shape
```

Grouping & Aggregation

```
df.groupby('col').size()         # Count by group
df.groupby('col').agg({'col2': 'mean', 'col3': 'sum'}) # Multi-column agg
df.groupby(['col1', 'col2']).agg(...) # Multi-level grouping
pd.crosstab(df['col1'], df['col2']) # Contingency table
```

Data Transformation

```
df['col'].apply(func)           # Apply function to column
df['col'].str.contains('pattern') # String search
df['col'].astype('category')    # Type conversion
pd.to_datetime(df['col'])       # Parse dates
df[df['col'] > threshold]       # Filtering
```

Visualization (Matplotlib/Seaborn)

```
df.plot()                      # Quick line plot
df['col'].hist(), df['col'].plot.box() # Histogram, boxplot
sns.heatmap(df.corr())         # Correlation heatmap
sns.countplot(data=df, x='col') # Bar chart
sns.scatterplot(data=df, x='col1', y='col2') # Scatter
sns.pairplot(df)              # Pairwise scatter matrix
```

SQL EDA

```
SELECT COUNT(*) FROM table;      -- Row count
SELECT COUNT(DISTINCT col) FROM table; -- Unique values
SELECT * FROM table LIMIT 5;     -- Preview
SELECT col, COUNT(*) FROM table GROUP BY col ORDER BY COUNT(*) DESC; -- Value counts
SELECT * FROM table WHERE col IS NULL; -- Find nulls
SELECT AVG(col), STDDEV(col), MIN(col), MAX(col) FROM table; -- Stats
```

Quick Start Pipeline (Copy-Paste)

```
df.info() df.describe(include='all') df.isnull().sum() df.duplicated().sum() df['target'].value_counts() # Class balance if classification
df.corr()
```

For AI/ML interviews and quick data exploration